

Design Reflection: Date changer app

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Reflection

In this homework, I implemented a date changer application using the Model–View–Controller (MVC) design pattern. The main design decision was to separate the date logic from the graphical user interface. The `DateModel` stores the current day, month, year, and the currently selected field. The `DateView` is responsible only for creating the Swing interface and displaying information. The `DateController` receives button events from the view, calls the appropriate methods in the model, and updates the view after each change.

The main functionalities of the application include displaying a date, selecting which field to modify, increasing or decreasing the selected field, and setting a selected field directly using a text input. The left and right arrow buttons are used to move between `DAY`, `MONTH`, and `YEAR`. The up and down buttons increase or decrease the selected value. The input field and `SET` button allow the user to enter a specific value for the selected field.

A key implementation detail is date validation. The model checks the number of days in each month and also handles leap years. For example, if the current date is January 31 and the month is changed to February, the day is adjusted to the maximum valid day of February. This prevents invalid dates such as February 31 from being displayed.

A strong point of my design is that the model is independent from Swing. This makes the logic easier to test separately before connecting it to the user interface. The program also follows clear class responsibilities: the model handles state and validation, the view handles presentation, and the controller handles user interaction. Overall, this homework helped me apply MVC to another interactive Java application and understand how state changes can be managed cleanly.